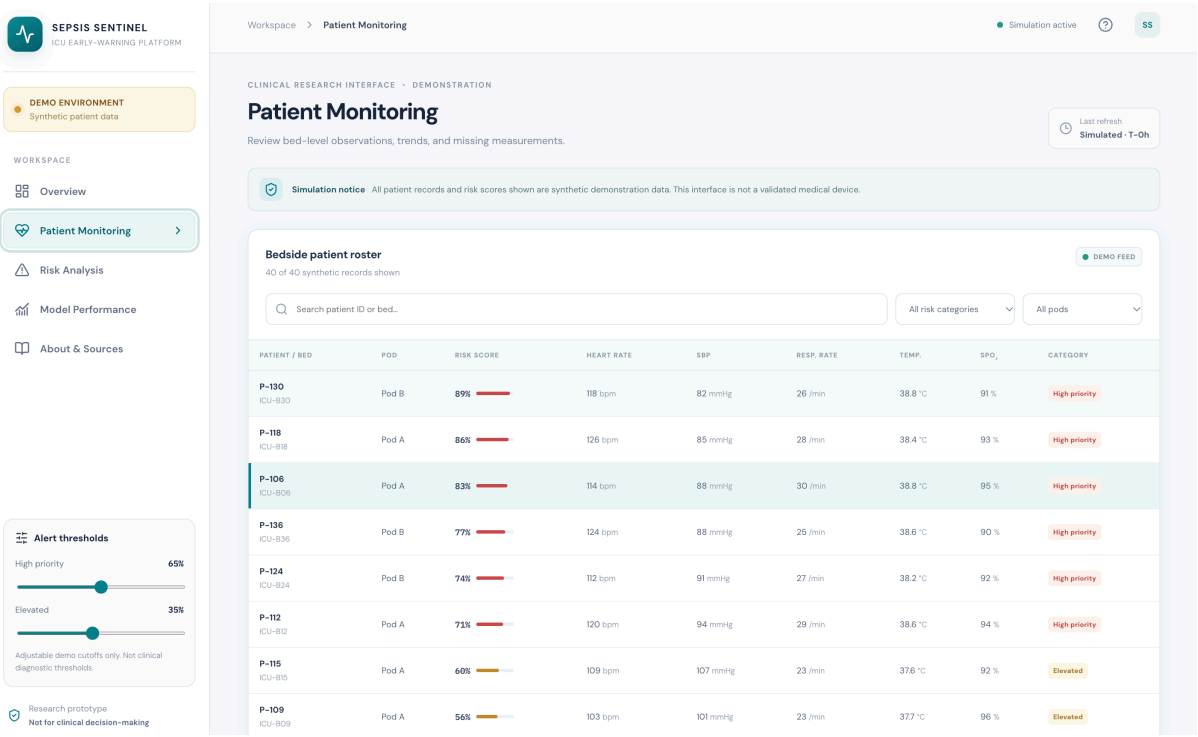
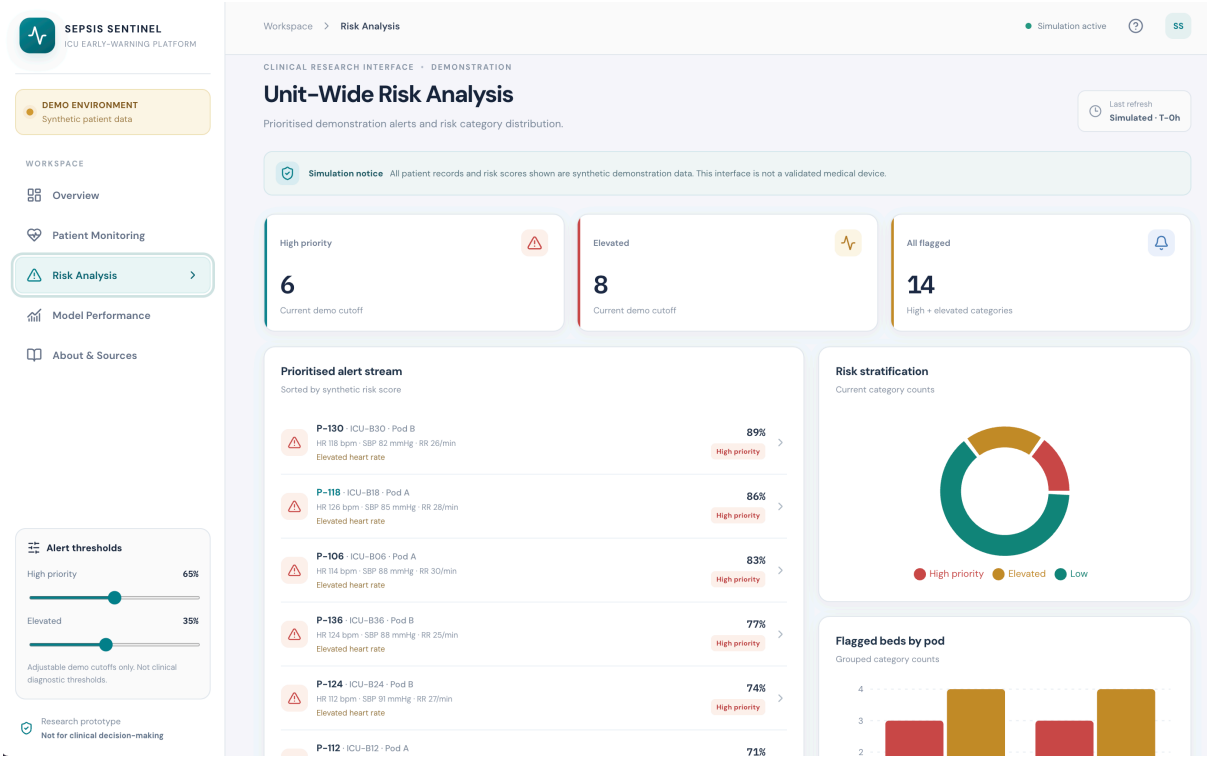
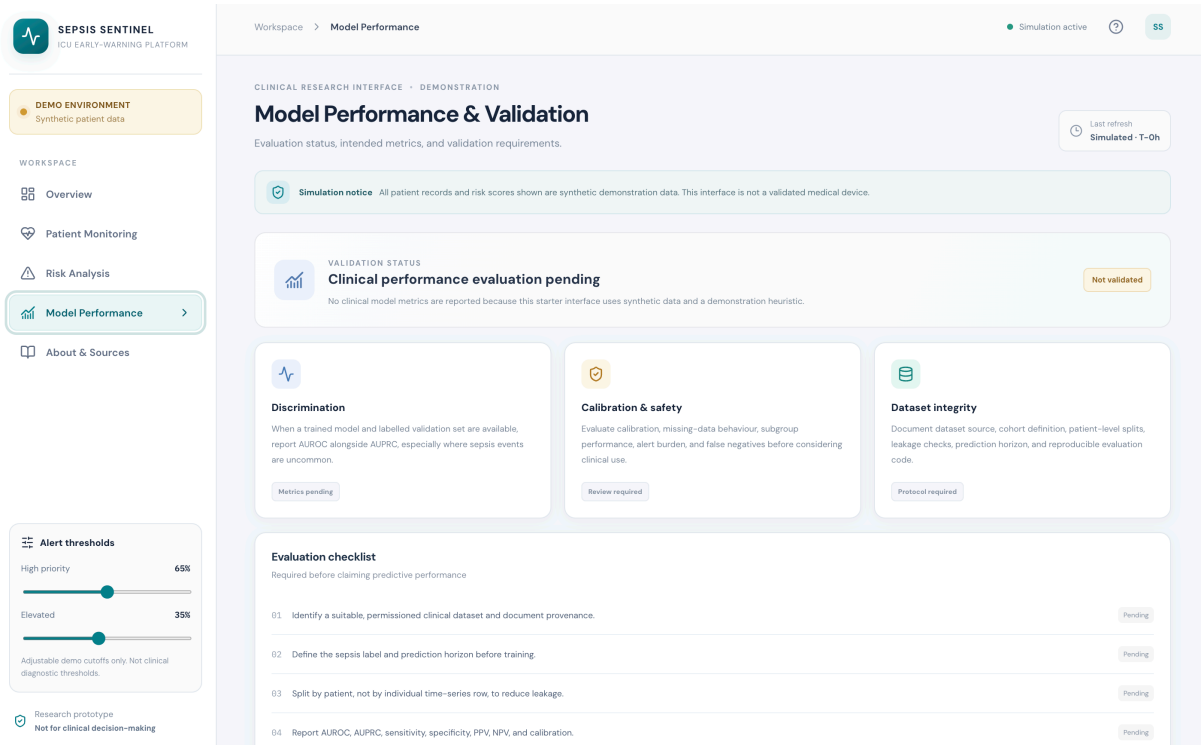
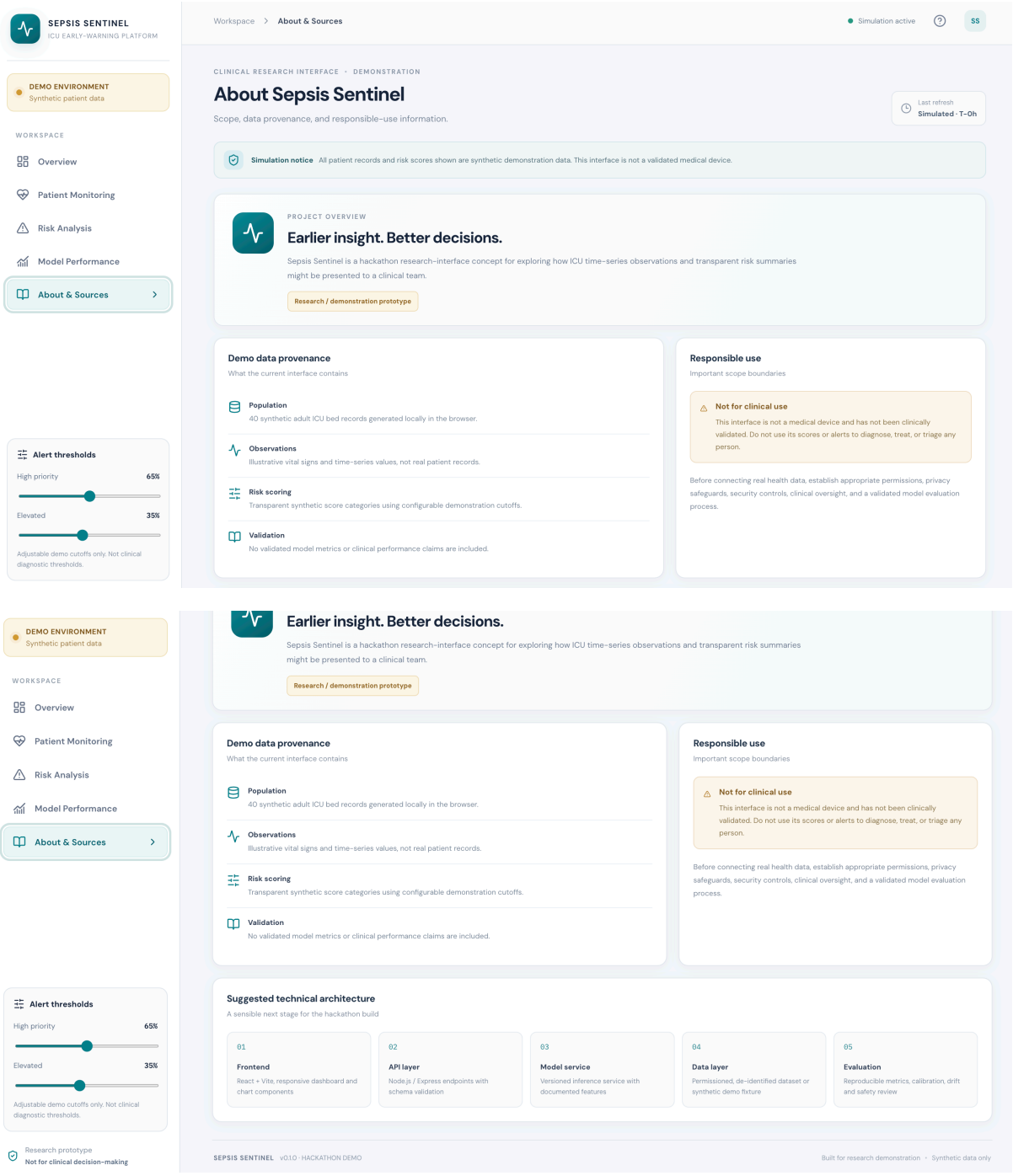

SEPSIS SENTINEL
ICU EARLY-WARNING PLATFORM
DEMO ENVIRONMENT
Synthetic patient data
WORKSPACE
Overview
Patient Monitoring
Risk Analysis
Model Performance
About & Sources
Alert thresholds
High priority 65%
Elevated 35%
Adjustable demo cutoffs only. Not clinical diagnostic thresholds.
Research prototype
Not for clinical decision-making
Workspace > Overview
Simulation active
SS
Patients monitored
40
Simulated ICU beds
High-priority flags
6
Above current demo threshold
Elevated risk flags
8
Requires review in demo flow
Data completeness
92%
Synthetic observation coverage
Risk distribution over time
Illustrative category counts across a simulated 24-hour window
24H WINDOW
32
24
16
8
0
00:00 02:00 04:00 06:00 08:00 10:00 12:00 14:00 16:00 18:00 20:00 22:00
High
Elevated
Low
Current risk stratification
Distribution of simulated beds by adjustable cutoff
LIVE FILTER
40
ICU beds
High priority 6 beds
Elevated 8 beds
Low 26 beds
Priority review queue
Highest demonstration scores first
View all alerts
P-130 · ICU-B30
118 bpm heart rate · SBP 82 mmHg
89%
High priority
P-118 · ICU-B18
126 bpm heart rate · SBP 85 mmHg
86%
High priority
P-106 · ICU-B06
114 bpm heart rate · SBP 88 mmHg
83%
High priority
Unit snapshot
Bed distribution by care pod
Pod A 20 beds
7 flagged beds
Pod B 20 beds
7 flagged beds
Scores are generated by a transparent demo heuristic—not a trained or validated prediction model.


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Workspace > Patient Monitoring
Simulation active
SS
CLINICAL RESEARCH INTERFACE · DEMONSTRATION
Patient Monitoring
Review bed-level observations, trends, and missing measurements.
Last refresh
Simulated · T-0h
Simulation notice All patient records and risk scores shown are synthetic demonstration data. This interface is not a validated medical device.
Bedside patient roster
40 of 40 synthetic records shown
DEMO FEED
Search patient ID or bed...
All risk categories
All pods
PATIENT / BED POD RISK SCORE HEART RATE SBP RESP. RATE TEMP. SPO2 CATEGORY
P-130 ICU-B30 Pod B 89% 118 bpm 82 mmHg 26 /min 38.8 °C 91 % High priority
P-118 ICU-B18 Pod A 86% 126 bpm 85 mmHg 28 /min 38.4 °C 93 % High priority
P-106 ICU-B06 Pod A 83% 114 bpm 88 mmHg 30 /min 38.8 °C 95 % High priority
P-136 ICU-B36 Pod B 77% 124 bpm 88 mmHg 25 /min 38.6 °C 90 % High priority
P-124 ICU-B24 Pod B 74% 112 bpm 91 mmHg 27 /min 38.2 °C 92 % High priority
P-112 ICU-B12 Pod A 71% 120 bpm 94 mmHg 29 /min 38.6 °C 94 % High priority
P-115 ICU-B15 Pod A 60% 109 bpm 107 mmHg 23 /min 37.6 °C 92 % Elevated
P-109 ICU-B09 Pod A 56% 103 bpm 101 mmHg 23 /min 37.7 °C 96 % Elevated


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Research prototype
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Workspace > Risk Analysis
Simulation active
SS
CLINICAL RESEARCH INTERFACE · DEMONSTRATION
Unit-Wide Risk Analysis
Prioritised demonstration alerts and risk category distribution.
Last refresh
Simulated · T-0h
Simulation notice All patient records and risk scores shown are synthetic demonstration data. This interface is not a validated medical device.
High priority
6
Current demo cutoff
Elevated
8
Current demo cutoff
All flagged
14
High + elevated categories
Prioritised alert stream
Sorted by synthetic risk score
P-130 · ICU-B30 · Pod B
HR 118 bpm · SBP 82 mmHg · RR 26/min
Elevated heart rate
89%
High priority
P-118 · ICU-B18 · Pod A
HR 126 bpm · SBP 85 mmHg · RR 28/min
Elevated heart rate
86%
High priority
P-106 · ICU-B06 · Pod A
HR 114 bpm · SBP 88 mmHg · RR 30/min
Elevated heart rate
83%
High priority
P-136 · ICU-B36 · Pod B
HR 124 bpm · SBP 88 mmHg · RR 25/min
Elevated heart rate
77%
High priority
P-124 · ICU-B24 · Pod B
HR 112 bpm · SBP 91 mmHg · RR 27/min
Elevated heart rate
74%
High priority
P-112 · ICU-B12 · Pod A
71%
Risk stratification
Current category counts
High priority
Elevated
Low
Flagged beds by pod
Grouped category counts
4
3
2


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Workspace > Model Performance
Simulation active
SS
CLINICAL RESEARCH INTERFACE · DEMONSTRATION
Model Performance & Validation
Evaluation status, intended metrics, and validation requirements.
Last refresh
Simulated · T-0h
Simulation notice All patient records and risk scores shown are synthetic demonstration data. This interface is not a validated medical device.
VALIDATION STATUS
Clinical performance evaluation pending
No clinical model metrics are reported because this starter interface uses synthetic data and a demonstration heuristic.
Not validated
Discrimination
When a trained model and labelled validation set are available, report AUROC alongside AUPRC, especially where sepsis events are uncommon.
Metrics pending
Calibration & safety
Evaluate calibration, missing-data behaviour, subgroup performance, alert burden, and false negatives before considering clinical use.
Review required
Dataset integrity
Document dataset source, cohort definition, patient-level splits, leakage checks, prediction horizon, and reproducible evaluation code.
Protocol required
Evaluation checklist
Required before claiming predictive performance
01 Identify a suitable, permissioned clinical dataset and document provenance.
Pending
02 Define the sepsis label and prediction horizon before training.
Pending
03 Split by patient, not by individual time-series row, to reduce leakage.
Pending
04 Report AUROC, AUPRC, sensitivity, specificity, PPV, NPV, and calibration.
Pending


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CLINICAL RESEARCH INTERFACE · DEMONSTRATION
About Sepsis Sentinel
Scope, data provenance, and responsible-use information.
Last refresh
Simulated · T-0h
Simulation notice All patient records and risk scores shown are synthetic demonstration data. This interface is not a validated medical device.
PROJECT OVERVIEW
Earlier insight. Better decisions.
Sepsis Sentinel is a hackathon research-interface concept for exploring how ICU time-series observations and transparent risk summaries might be presented to a clinical team.
Research / demonstration prototype
Demo data provenance
What the current interface contains
Population
40 synthetic adult ICU bed records generated locally in the browser.
Observations
Illustrative vital signs and time-series values, not real patient records.
Risk scoring
Transparent synthetic score categories using configurable demonstration cutoffs.
Validation
No validated model metrics or clinical performance claims are included.
Responsible use
Important scope boundaries
Not for clinical use
This interface is not a medical device and has not been clinically validated. Do not use its scores or alerts to diagnose, treat, or triage any person.
Before connecting real health data, establish appropriate permissions, privacy safeguards, security controls, clinical oversight, and a validated model evaluation process.

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Suggested technical architecture
A sensible next stage for the hackathon build
01
Frontend
React + Vite, responsive dashboard and chart components
02
API layer
Node.js / Express endpoints with schema validation
03
Model service
Versioned inference service with documented features
04
Data layer
Permissioned, de-identified dataset or synthetic demo fixture
05
Evaluation
Reproducible metrics, calibration, drift and safety review
SEPSIS SENTINEL v0.1.0 · HACKATHON DEMO
Built for research demonstration · Synthetic data only